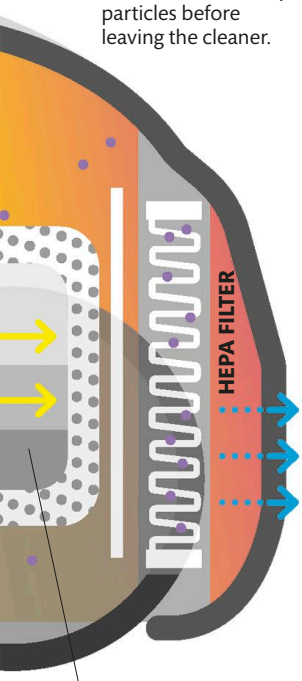


4 Air expelled
The air cools the motor as it passes by. It then travels through a HEPA filter to remove microscopic particles before leaving the cleaner.

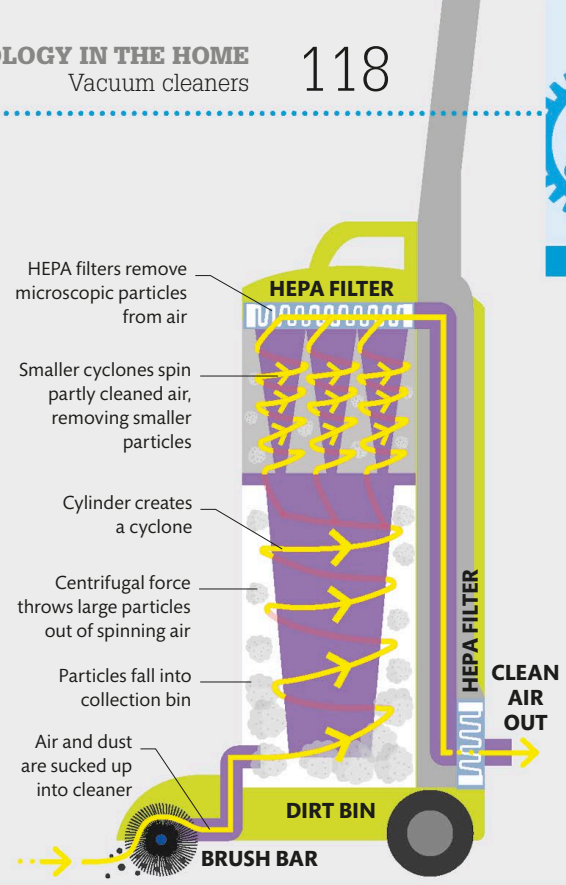


Motor spins fan at great speed, usually at hundreds or thousands of revolutions per minute

Cyclonic vacuum cleaners

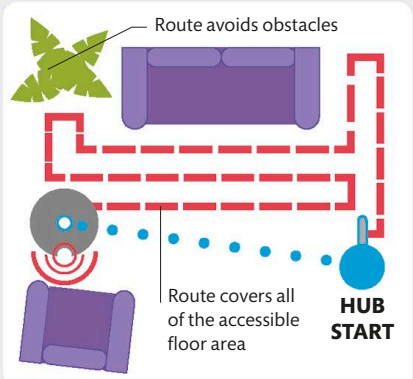
This type of cleaner dispenses with a dust bag and does not suffer from filters getting clogged with large or medium particles during cleaning. It relies on vortices of air (known as cyclones) spinning the air to fling particles out of the airstream. HEPA filters remove tiny particles from the air. They should be cleaned or replaced every 6 months.

THE **MOTORS** OF SOME **CYCLONIC VACUUM CLEANERS** SPIN UP TO **120,000 REVOLUTIONS PER MINUTE**



Robotic cleaners

These mobile robots, driven by electric motors, navigate themselves around a living space as they clean floors. A package of sensors enables the robot to measure how far it has traveled and to detect obstacles. They also include cliff sensors, which spot sudden drops ahead such as stairs. After a cleaning session, the robot can guide itself to its charging station to recharge its batteries.



Navigation
The robot's microprocessor-based controller runs software that plots a route around a room or rooms, ensuring total cleaning coverage. The robot keeps track of its location and can replot its route should an obstacle bar its way.

